

Mirror Descent

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1 From Gradient Descent to Mirror Descent

1.1 Gradient Descent from a Proximal Perspective

We consider a simple optimization problem

$$\min_{\mathbf{x}} f(\mathbf{x}), \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^d$ and $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is a differentiable objective function. To solve the problem numerically, one of the most classic and fundamental methods is gradient descent. Starting from an initial point $\mathbf{x}_0$, it iteratively updates the solution by moving in the negative gradient direction:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_{t+1} \nabla f(\mathbf{x}_t), \quad (2)$$

where $\nabla f(\mathbf{x}_t) \in \mathbb{R}^d$ is the gradient at $\mathbf{x}_t$ and $\eta_t > 0$ is the step size (or learning rate).

This procedure is commonly interpreted as iteratively moving in the direction that decreases the objective function most rapidly, namely, the negative gradient direction. The parameter η_t controls the step size at the t -th iteration.

However, gradient descent can also be viewed from another perspective. From (2), we note that

$$\begin{aligned} \mathbf{x}_{t+1} = \mathbf{x}_t - \eta_{t+1} \nabla f(\mathbf{x}_t) &= \operatorname{argmin}_{\mathbf{x}} \left\{ \eta_{t+1} \langle \nabla f(\mathbf{x}_t), \mathbf{x} - \mathbf{x}_t \rangle + \frac{1}{2} \|\mathbf{x} - \mathbf{x}_t\|_2^2 \right\} \\ &= \operatorname{prox}_{\eta_{t+1} \langle \nabla f(\mathbf{x}_t), \mathbf{x} - \mathbf{x}_t \rangle}(\mathbf{x}_t), \end{aligned} \quad (3)$$

where $\|\cdot\|_2$ is the ℓ_2 norm, $\langle \cdot, \cdot \rangle$ denotes the inner product, and $\operatorname{prox}_h(\mathbf{x})$ is the **proximal operator** defined as

$$\operatorname{prox}_h(\mathbf{x}) = \operatorname{argmin}_{\mathbf{u}} \left\{ h(\mathbf{u}) + \frac{1}{2} \|\mathbf{u} - \mathbf{x}\|_2^2 \right\}. \quad (4)$$

In this formulation, gradient descent can be intuitively interpreted as finding the next point $\mathbf{x}_{t+1}$ that not only decreases the linear approximation of objective f , but also *remains close to the current point $\mathbf{x}_t$* .

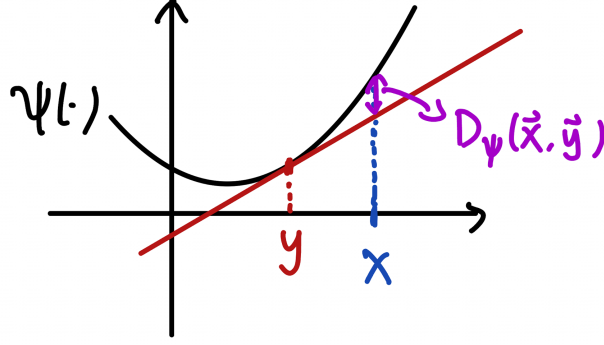


Figure 1: Visualization of Bregman divergence

1.2 Bregman Divergence

Based on the formulation and intuition above, we can naturally extend gradient descent to mirror descent. We note that in gradient descent, the closeness between $\mathbf{x}_{t+1}$ and $\mathbf{x}_t$ is measured by ℓ_2 norm. A natural way to generalize this framework is to replace the squared Euclidean distance with a more general divergence measure. In mirror descent, the distance is measured by **Bregman divergence**, which is defined as

$$D_\psi(\mathbf{x}, \mathbf{y}) = \psi(\mathbf{x}) - \psi(\mathbf{y}) - \langle \nabla \psi(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle, \quad (5)$$

where $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ is a differentiable strictly convex function. Sometimes ψ is called the **distance-generating function** since it generates the distance measure. Figure 1 provides an intuitive visualization of Bregman divergence, which measures the difference between actual function value $\psi(\mathbf{x})$ and the linear approximation $\psi(\mathbf{y}) + \langle \nabla \psi(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$ at point $\mathbf{y}$.

We can also conclude the basic properties of Bregman divergence as follows.

Proposition 1 (Basic properties of Bregman divergence).

- (a) **Non-negativity:** $D_\psi(\mathbf{x}, \mathbf{y}) \geq 0$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$.
- (b) **Positivity:** $D_\psi(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$.
- (c) **Non-commutativity:** In general, $D_\psi(\mathbf{x}, \mathbf{y}) \neq D_\psi(\mathbf{y}, \mathbf{x})$.

1.3 Mirror Descent

We replace the Euclidean distance in (3) with Bregman divergence defined in (5), which yields

$$\begin{aligned} \mathbf{x}_{t+1} &= \operatorname{argmin}_{\mathbf{x}} \{ \eta_{t+1} \langle \nabla f(\mathbf{x}_t), \mathbf{x} - \mathbf{x}_t \rangle + D_\psi(\mathbf{x}, \mathbf{x}_t) \} \\ &= \operatorname{argmin}_{\mathbf{x}} \{ \eta_{t+1} \langle \nabla f(\mathbf{x}_t), \mathbf{x} - \mathbf{x}_t \rangle + \psi(\mathbf{x}) - \psi(\mathbf{x}_t) - \langle \nabla \psi(\mathbf{x}_t), \mathbf{x} - \mathbf{x}_t \rangle \} \\ &\triangleq \operatorname{argmin}_{\mathbf{x}} h(\mathbf{x}). \end{aligned} \quad (6)$$

Hence, the next iteration point $\mathbf{x}_{t+1}$ is such $\mathbf{x}$ that minimizes $h(\mathbf{x})$. We note that $h(\mathbf{x})$ is strictly convex, since for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$, we have

$$h(\mathbf{x}_2) - h(\mathbf{x}_1) - \langle \nabla h(\mathbf{x}_1), \mathbf{x}_2 - \mathbf{x}_1 \rangle = \psi(\mathbf{x}_2) - \psi(\mathbf{x}_1) - \langle \nabla \psi(\mathbf{x}_1), \mathbf{x}_2 - \mathbf{x}_1 \rangle \geq 0, \quad (7)$$

and the equation holds if and only if $\mathbf{x}_1 = \mathbf{x}_2$ by Proposition 1 (b). Therefore, $\mathbf{x}_{t+1}$ satisfies

$$\nabla h(\mathbf{x}_{t+1}) = \eta_{t+1} \nabla f(\mathbf{x}_t) + \nabla \psi(\mathbf{x}_{t+1}) - \nabla \psi(\mathbf{x}_t) = \mathbf{0}, \quad (8)$$

which is equivalent to

$$\nabla \psi(\mathbf{x}_{t+1}) = \nabla \psi(\mathbf{x}_t) - \eta_{t+1} \nabla f(\mathbf{x}_t). \quad (9)$$

From (9) we note that we are actually updating $\nabla \psi(\mathbf{x}_t)$ with a gradient descent procedure with the gradient $\nabla f(\mathbf{x}_t)$. Hence, mirror descent can be interpreted as performing gradient descent in the dual space induced by the **mirror map** $\nabla \psi$. Specifically, we first map the primal iterate $\mathbf{x}_t$ to the dual variable

$$\mathbf{z}_t \triangleq \nabla \psi(\mathbf{x}_t), \quad (10)$$

perform a gradient update in the dual space to obtain

$$\mathbf{z}_{t+1} = \mathbf{z}_t - \eta_{t+1} \nabla f(\mathbf{x}_t), \quad (11)$$

and then map back to the primal space via

$$\mathbf{x}_{t+1} = (\nabla \psi)^{-1}(\mathbf{z}_{t+1}), \quad (12)$$

assuming that $\nabla \psi$ is invertible. Additionally, it should be noted that gradient descent is a special case of mirror descent by choosing $\psi(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|_2^2$.

The procedure of mirror descent is summarized in Algorithm 1.

Algorithm 1 Mirror Descent

Require: Initial point $\mathbf{x}_0 \in \mathcal{X}$, mirror map ψ , step sizes $\{\eta_t\}_{t \geq 1}$, objective function f

- 1: **for** $t = 0, 1, \dots, T - 1$ **do**
 - 2: Map to the dual space: $\mathbf{z}_t = \nabla \psi(\mathbf{x}_t)$
 - 3: Perform a gradient update in the dual space $\mathbf{z}_{t+1} = \mathbf{z}_t - \eta_{t+1} \nabla f(\mathbf{x}_t)$
 - 4: Map back to the primal space $\mathbf{x}_{t+1} = (\nabla \psi)^{-1}(\mathbf{z}_{t+1})$
 - 5: **end for**
 - 6: **return** $\mathbf{x}_T$
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2 Example: Optimization on the Probability Simplex

In this section, we present several examples to show how mirror descent can better exploit the geometry of certain optimization problems than standard gradient descent. Optimization

on the probability simplex is perhaps the most classic example of mirror descent. In this setting, our goal is

$$\begin{aligned} \min f(\mathbf{x}), \\ \text{s.t. } \mathbf{x} \in \Delta_d, \end{aligned} \tag{13}$$

where $\Delta_d = \left\{ \mathbf{x} \in \mathbb{R}_+^d, \sum_{i=1}^d x_i = 1 \right\}$ is the probability simplex on $\mathbb{R}^d$. In gradient descent, since $\mathbf{x}$ is restricted to the probability simplex, we have to project the point onto the simplex after performing the gradient descent step, which is

$$\mathbf{x}_{t+1} = \text{proj}_{\Delta_d}(\mathbf{x}_t - \eta_{t+1} \nabla f(\mathbf{x}_t)), \tag{14}$$

where $\text{proj}_A(\mathbf{x}) \triangleq \text{argmin}_{\mathbf{u} \in A} \{\|\mathbf{u} - \mathbf{x}\|_2\}$ is the projection operator. This approach is somewhat unnatural, since a gradient step may leave the feasible region, requiring an additional projection onto the simplex. Besides, the Euclidean distance does not adequately capture the geometry of probability distributions.

In mirror descent, we can choose $\psi(\mathbf{x}) = \sum_{i=1}^d x_i \log x_i$, which is known as the negative entropy mirror map. The corresponding Bregman divergence is $D_\psi(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^d x_i \log \frac{x_i}{y_i}$, namely the **Kullback–Leibler (KL) divergence**. The update of mirror descent is

$$x_{t+1,i} = \frac{x_{t,i} \exp(-\eta_{t+1} \nabla f(x_t)_i)}{\sum_j x_{t,j} \exp(-\eta_{t+1} \nabla f(x_t)_j)}, \tag{15}$$

where each element $x_{t,i}$ is first updated via multiplying $\exp(-\eta_{t+1} \nabla f(x_t)_i)$, then normalized by dividing $\sum_j x_{t,j} \exp(-\eta_{t+1} \nabla f(x_t)_j)$. The procedure is more natural since the multiplication update guarantees $x_{t+1,i} > 0$ for all i and is more suitable for probabilities. Moreover, the simplex constraint is enforced automatically through the normalization constant in the exponentiated gradient update, avoiding the need for an explicit projection step.